

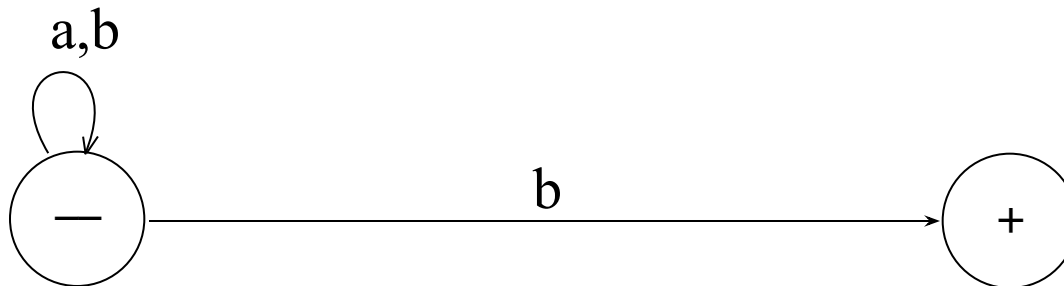
Recap lecture 8



- **TG definition, Examples: accepting all strings, accepting none, starting with b, not ending in b, containing aa, containing aa or bb**

Task Solution

- Build a TG accepting the language L of strings, defined over $\Sigma=\{a, b\}$, **ending in b.**
- **Solution** The language L may be expressed by RE $(a + b)^*b$, may be accepted by the following TG



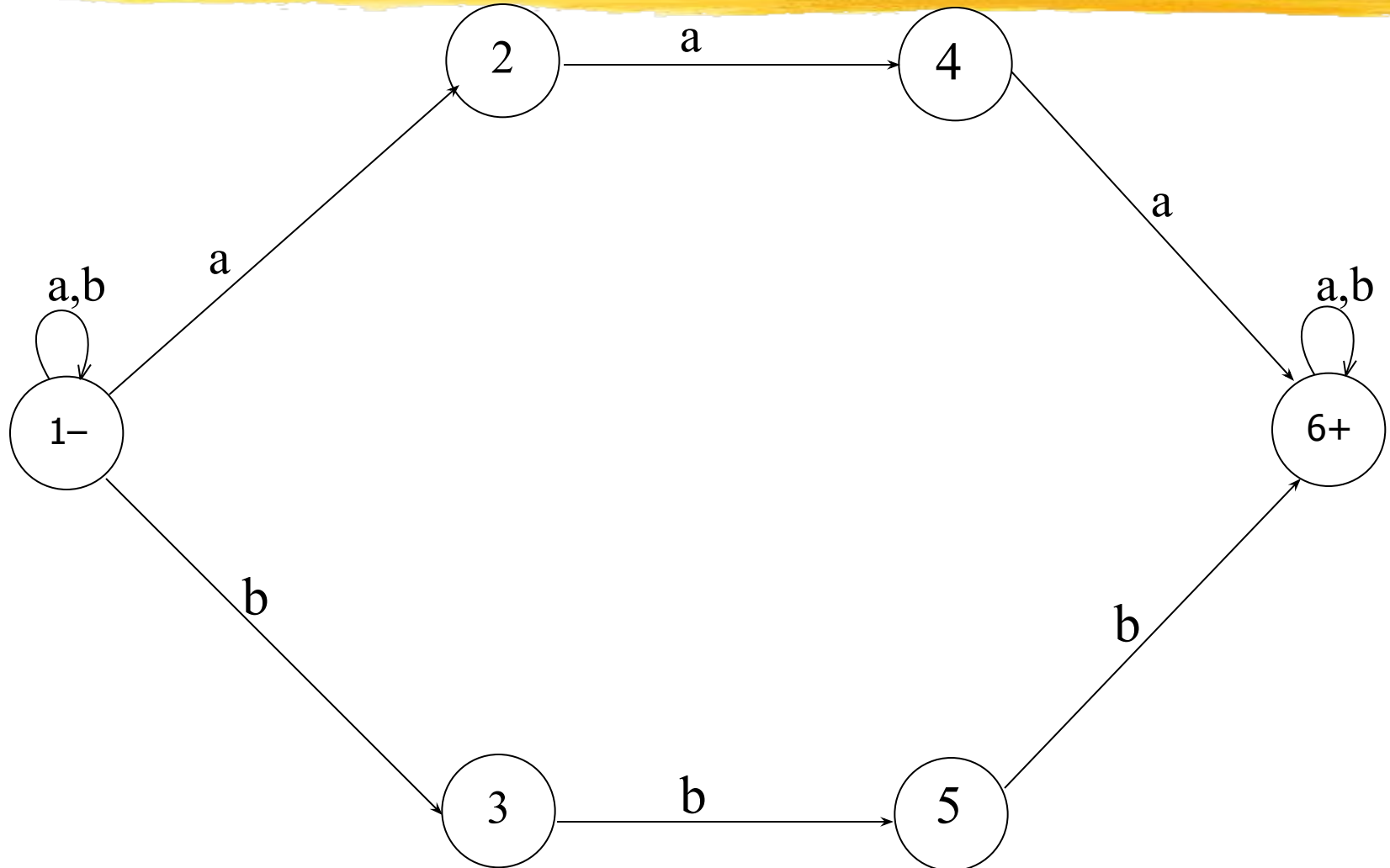
Example

Consider the language L of strings, defined over $\Sigma = \{a, b\}$, **having triple a or triple b.** The language L may be expressed by RE

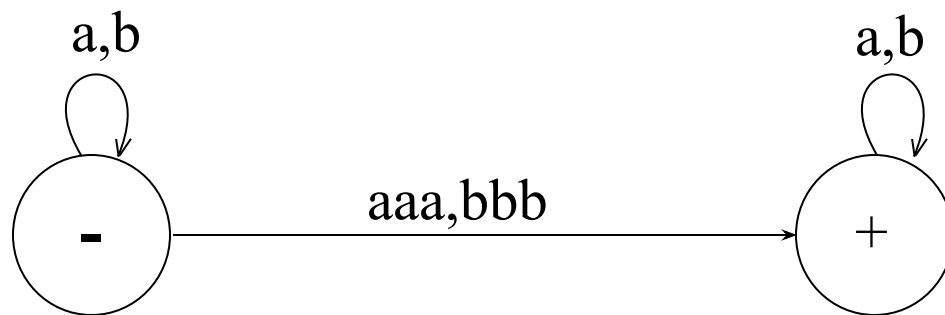
$$(a+b)^* (aaa + bbb) (a+b)^*$$

This language may be accepted by the following TG

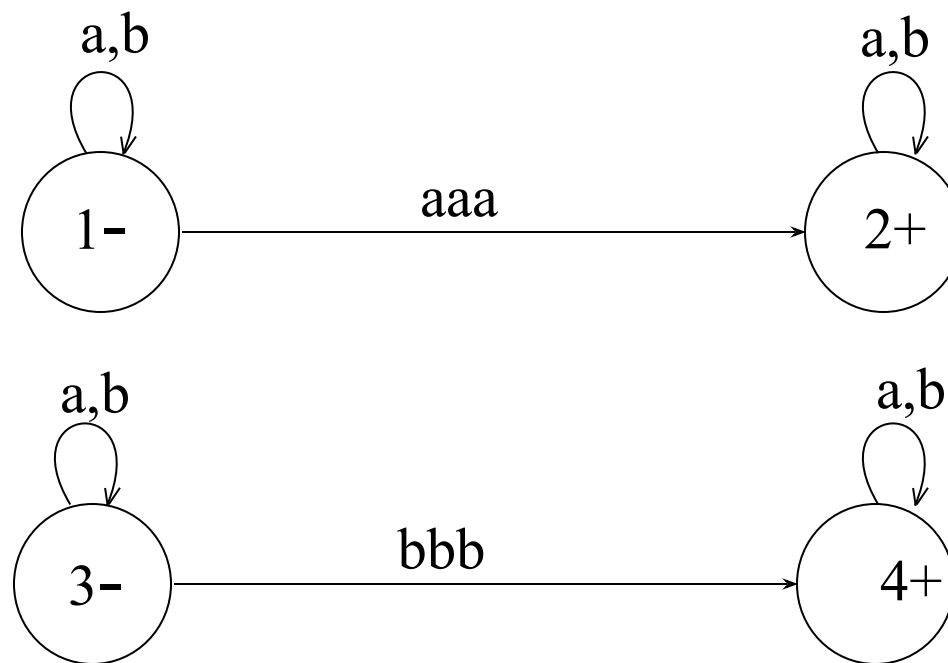
Example Continued ...



OR



OR



Example

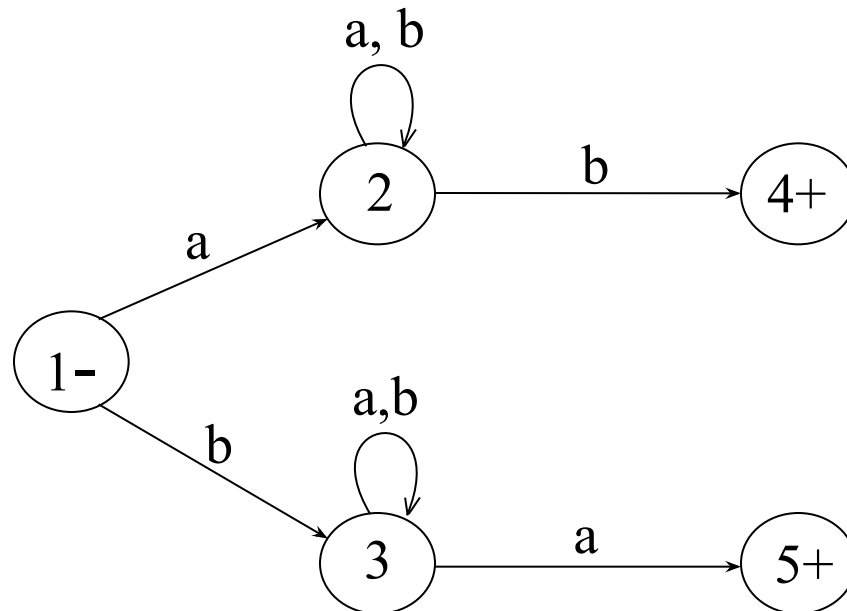


Consider the language L of strings, defined over $\Sigma = \{a, b\}$, **beginning and ending in different letters.**

The language L may be expressed by RE
 $a(a + b)^*b + b(a + b)^*a$

The language L may be accepted by the following TG

Example continued ...



Example

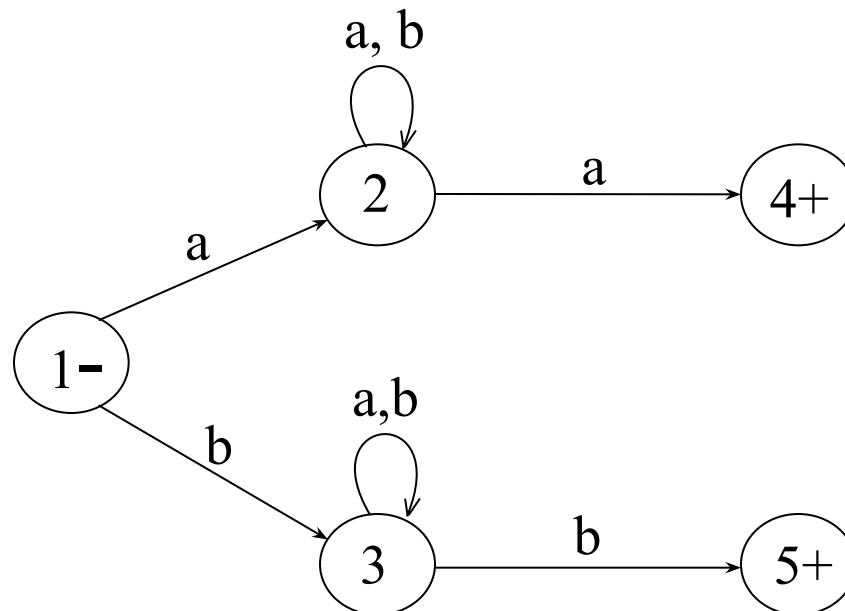
- Consider the Language L of strings **of length two or more**, defined over $\Sigma = \{a, b\}$, **beginning with and ending in same letters.**

The language L may be expressed by the following regular expression

$$a(a + b)^*a + b(a + b)^*b$$

This language may be accepted by the following TG

Example Continued ...



Task

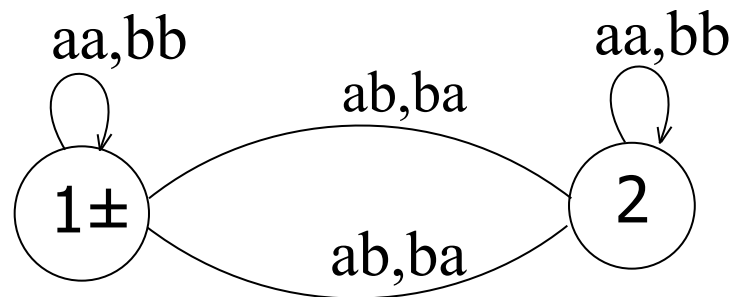


Build a TG accepting the language L of strings, defined over $\Sigma = \{a, b\}$, **beginning with and ending in the same letters.**

Example

- Consider the **EVEN-EVEN** language, defined over $\Sigma = \{a, b\}$. As discussed earlier that **EVEN-EVEN** language can be expressed by a regular expression
 $(aa+bb+(ab+ba)(aa+bb)^*(ab+ba))^*$
The language **EVEN-EVEN** may be accepted by the following TG

Example continued ...



Example

- Consider the language L , defined over $\Sigma = \{a, b\}$, in which **a's occur only in even clumps and that ends in three or more b's**. The language L can be expressed by its regular expression

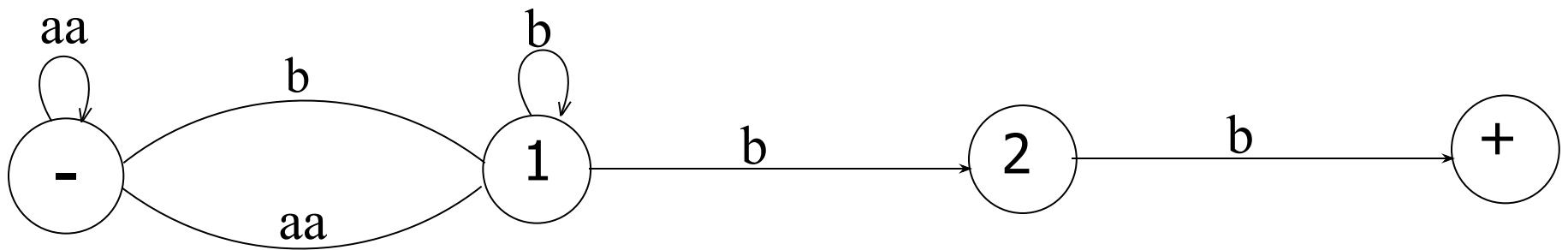
$(aa)^*b(b^* + (aa(aa)^*b)^*)bb$

OR

$(aa)^*b(b^* + ((aa)^+b)^*)bb$

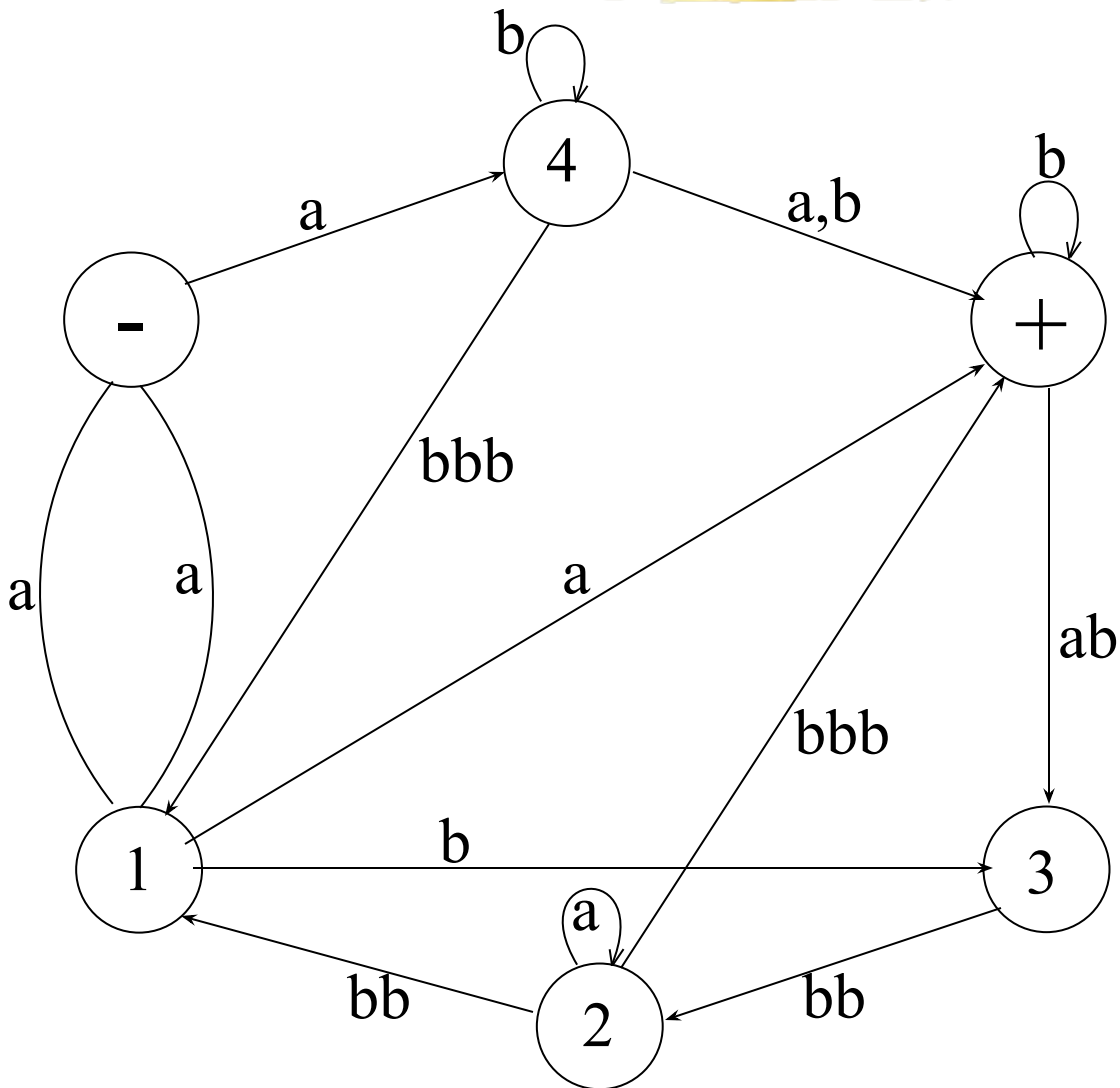
The language L may be accepted by the following TG

Example Continued ...



**Example:
following TG**

Consider the



Generalized Transition Graphs

A generalized transition graph (GTG) is a collection of three things

- 1) Finite number of states, at least one of which is start state and some (maybe none) final states.
- 2) Finite set of input letters (Σ) from which input strings are formed.
- 3) Directed edges connecting some pair of states labeled with regular expression.

It may be noted that in GTG, the labels of transition edges are corresponding regular expressions

Summing Up



- **TGs accepting the languages: containing aaa or bbb, beginning and ending in different letters, beginning and ending in same letters, EVEN-EVEN, a's occur in even clumps and ends in three or more b's, example showing different paths traced by one string, Definition of GTG**